The Reggae Shark Protocol (v1.2.0 — Red-Team Tested Edition): Formal Typestate Proofs (38 Lean 4 Theorems), 4D Soft-Wall ODE Governance, Bandlimited LRSC Compression, and Ed25519-Signed Merkle-DAG Provenance

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Abstract

The Reggae Shark Protocol (v1.2.0) is a verifiable execution kernel and multi-cloud AI gateway that binds agent prompt dispatches to four mathematical and cryptographic verification layers: (1) an append-only, SHA-256 hash-chained and RFC 8032 Ed25519-signed **IntentSeal Verifiable Research Kernel (VRK v0.3.1)** ledger enforcing the `intentseal-vrk-claim-v2` schema, prompt/response SHA-256 commitments, on-disk evidence anchor checks, and an N-way commutative, associative, and idempotent Merkle-DAG join-semilattice for distributed split-brain reconciliation; (1B) a thread-synchronized **Request-Lifecycle Typestate and Worker-Epoch Isolation** state machine (`TypestateLifecycle.lean`, 12 zero-`sorry` theorems) that rules out partial-initialization payload dereferencing and cross-epoch worker-slot reads across multi-step event traces within the modelled state machine; (2) a bounded-LRU per-tenant **4D Soft-Wall ODE Cognitive Dynamics Governor** $(\gamma, H_{\text{obs}}, \dot{I}, \Psi)$ backed by 17 Lean 4 4D boundary theorems (`ExistenceVerification.lean`), 9 3D soft-wall theorems (`BoundaryVerification.lean`), and a subnormal-safe Patankar–Euler interval integrator; and (3) a **Bandlimited Low-Rank Spectral Context Memory (LRSC)** operator $(M = 99, K = 11)$ with dual compressed/uncompressed SHA-256 sealing and a bounded-domain local attention sensitivity bound.

1 Four-Layer Verification Architecture & Provenance (38 Lean 4 Theorems)

Every request submitted to `MultiProviderGateway.process_request` or the local loopback HTTP proxy (`POST http://127.0.0.1:8000/v1/execute`) traverses four deterministic verification gates:

Layer	Module / Artifact	Enforced Mathematical & Cryptographic Invariant
Layer 1	`shark_protocol/kernel.py`	**Ed25519-Signed VRK v0.3.1 Chain & N-Way Merkle-DAG Join:** Append-only SHA-256 chain anchored at trust root `f010a99d...05a0d0f` (10.5281/zenodo.23018459), committing to `prompt_sha256`, `response_sha256`, on-disk `evidence_root`, Ed25519 signatures, and verified `parent_hashes` membership.
Layer 1B	`lean/TypestateLifecycle.lean`	**Typestate & Worker-Epoch Automaton (12 Theorems) + Python RLock:** 6-state request automaton (`UNINIT` → `PARTIAL_INIT` → `SEALED_ACTIVE` → `SHUTDOWN_CLEAN` → `ZEROED_FREED`) with absorbing `ZEROED_FREED`, inductive trace invariants (`runTrace`), and Python `guard._lock` → `slot._lock` synchronization.
Layer 2	`lean/ExistenceVerification.lean` `lean/BoundaryVerification.lean`	**4D Soft-Wall ODE Governor ($\mathcal{B}_4$, 26 Theorems):** Bounded-LRU per-tenant adaptive RK4 and Patankar–Euler interval integration of $(\gamma, H_{\text{obs}}, \dot{I}, \Psi)$ inside $\mathcal{B}_4 = [0,10] \times [0,100] \times [0, \dot{I}_{\max,4D}] \times [0,1]$ with Python float64 floor 2.225×10^{-308}.
Layer 3	`LRSCCompressor` (`kernel.py`)	**Bandlimited LRSC Odd-Ring Projection (11/99 = 1/9):** Projects context onto the $M = 99$ odd-ring filter V_{ring} ($\sigma_{\min}(S_{11}) = 0.181646 > 0$, 10.17605/OSF.IO/NM5BW), evaluates a bounded-domain local sensitivity proxy [5], and binds distinct compressed/uncompressed SHA-256 digests.
Layer 4	`shark_protocol/gateway.py`	**Fail-Closed Multi-Tenant Gateway & ThreadingHTTPServer:** Enforces waking basal drive $\mathcal{P}_0 \geq 1.0 \gg 10/99$, bounded tenant LRU (`max_tenants=128`), header-only API keys (`x-goog-api-key`), HTTP 413 payload caps, and fail-closed live dispatch.

Table 1: Four-layer verification architecture of The Reggae Shark Protocol (v1.2.0).

Canonical Lean 4 Module Provenance (38 Zero-`sorry` Theorems): The 24-file release archive contains 23 hash-verified files plus `MANIFEST_SHA256.txt`. Specifically, `lean/BoundaryVerification.lean` (`SHA-256: 49229b61...e417`, 9 theorems) is byte-for-byte identical to the canonical module in Physics v1.5 Appendix A (10.5281/zenodo.23534449), and `lean/ExistenceVerification.lean` (`SHA-256: eec94868...bcae`, 17 theorems) is byte-for-byte identical to the canonical module in *The Theory of Bounded Existence* v1.0.0 (10.5281/zenodo.23608244). Together with the 12 theorems in `lean/TypestateLifecycle.lean` (each verified with `#print axioms`), the package compiles **38 zero-`sorry` Lean 4 theorems**.

2 What Is and Is Not Modelled (Epistemic Claim Firewall)

Epistemic Standard (`Build. Attack. Falsify. Preserve.`): Every theorem and runtime guard in this repository is bounded by explicit domain assumptions and `scope.does_not_imply` declarations.

- **Application-Layer Typestate vs. Native Runtimes:** `lean/TypestateLifecycle.lean` proves multi-step trace properties of the formal state-transition model $\delta : \mathcal{S} \times \Sigma \to \mathcal{S}$, while `RequestSlot` and `WorkerEpochGuard` enforce those invariants and `threading.RLock` ordering in Python. They are *not* a patch to native C/PHP interpreters or an OS ring-0 memory boundary.
- **Real Arithmetic in Lean vs. IEEE-754 Runtime Guards:** Lean 4 Theorems 6, 7, 9, and 10 are proved over the mathematical real field $\mathbb{R}$; IEEE-754 float64 subnormal clamping at `FLOAT64_NORMAL_MIN` and numerical Grönwall floor checks are enforced in Python runtime code.
- **Offline Simulation vs. External Cloud APIs:** Automated CI exercises and verifies the deterministic local pipeline (`VRK_OFFLINE_SEALED_SIMULATION`) and local `127.0.0.1` HTTP loopback proxy. Live external cloud dispatch (`LIVE_API_DISPATCH`) requires user API keys and fails closed with `GatewayDispatchError` if an upstream call fails.

3 Layer 2 & Layer 3 Mathematical Equations

The Layer 2 governor evolves $x(t) = (\gamma(t), H_{\text{obs}}(t), \dot{I}(t), \Psi(t))$ under the smooth (C^∞) soft-wall system [3]:

$$\frac{d\gamma}{dt} = a\dot{I}\mathcal{E}_{\text{avail}}\left(1 - \frac{\gamma}{\gamma_{\max}}\right) - b\gamma, \tag{1}$$

$$\frac{dH_{\text{obs}}}{dt} = c\gamma\,(H_{\max} - H_{\text{obs}}) - \dot{I}_{\text{erase}}\tanh\!\left(\frac{H_{\text{obs}}}{\epsilon_H}\right), \tag{2}$$

$$\frac{d\dot{I}}{dt} = e\tanh\!\left(\frac{\dot{I}}{\epsilon_I}\right)\frac{dH_{\text{obs}}}{dt} + f\mathcal{P}_{\text{eff}}(\gamma,\Psi) - g\dot{I}, \tag{3}$$

$$\frac{d\Psi}{dt} = \kappa_{\text{meta}}R_{\text{self}}(x)\,(1-\Psi) - \mu_\Psi\Psi, \tag{4}$$

where $\mathcal{P}_{\text{eff}}(\gamma,\Psi) = \mathcal{P}_0 + \lambda_{\text{sim}}r_{\text{LRSC}}\gamma + \lambda_{\text{lucid}}\Psi$ and $R_{\text{self}}(x) = \text{fb}(x)^2/(\text{fb}(x)^2 + (f\mathcal{P}_0)^2 + \sigma_0^2)$ with $\text{fb}(x) = e\tanh(\dot{I}/\epsilon_I)(dH_{\text{obs}}/dt)$.

4 Five Core Verification Guards & Five Edge-Case Probes (30/30 Tests)

1. **Guard 1 & Probe A (Patankar–Euler Box Invariance & Python Subnormal Floor):** `step_ode_patankar_interval` updates each coordinate via the cleared-denominator scheme

$$x_{k+1} = \frac{x_k + \Delta t\,P(x_k)\,x_{\max}}{1 + \Delta t\,(P(x_k) + D(x_k))}, \qquad P(x_k) \geq 0,\ D(x_k) \geq 0,\ 1 + \Delta t(P+D) \geq 1, \tag{5}$$

   paired with 10^{-12} Taylor cutoffs and a Python runtime float64 floor `FLOAT64_NORMAL_MIN` $= 2.225 \times 10^{-308}$ whenever $x_k > 0$ or $P(x_k) > 0$. Lean 4 Theorems `patankar_euler_step_in_box` and `patankar_strict_positivity_floor` prove the real-arithmetic bounds over $\mathbb{R}$ (`test_15`, `test_21`).
2. **Guard 2 & Probe B (Pointwise Decay Bound, Bounded Tenant LRU & Waking Drive Floor):** Lean 4 Theorem `psi_decay_floor_bound` proves the pointwise derivative inequality $d\Psi/dt \geq -\mu_\Psi\Psi$, which by the classical Grönwall lemma yields $\Psi(t+\Delta t) \geq \Psi(t)e^{-\mu_\Psi\Delta t}$. `MultiProviderGateway` bounds per-tenant engines in an `OrderedDict` LRU cache (`max_tenants=128`) and enforces $\mathcal{P}_0 \geq 1.0 \gg 10/99$ (`test_16`, `test_22`, `test_29`).
3. **Guard 3 & Probe C (Prompt-Committed Bandlimited LRSC Witness & Local Sensitivity Proxy):** `LRSCCompressor.compress` injects a baseline anchor on slot 0 to eliminate scalar-multiple normalization collisions (distinguishing single-byte and constant-byte strings in `test_27`), projects onto V_{ring} ($\sigma_{\min}(S_{11}) = 0.18164624 > 0$ [4]), evaluates the local bounded-domain sensitivity proxy $L_{\text{local}}(L_W, d_k)\eta_\perp(\tilde{x})$ on $\|\tilde{x}\|_2 \leq 1$ [5] (Theorem `lrsc_attention_lipschitz_error_bound`), and seals `prompt_sha256`, `response_sha256`, `lrsc_sha256`, and `uncompressed_ring_sha256` (`test_17`, `test_23`, `test_27`).
4. **Guard 4 & Probe D (Inductive Typestate Trace Safety, Absorbing `ZEROED_FREED` & RLock Order):** Lean 4 Theorems 1–3 and 5 prove by induction over arbitrary event lists `evs : List LifecycleEvent` that `ZeroedFreed` is strictly absorbing and that `{AbortedPoisoned, ZeroedFreed}` never reaches `SealedActive` or `ShutdownClean`, while Python enforces `WorkerEpochGuard._lock` → `RequestSlot._lock` (`test_18`, `test_24`, `test_30`).
5. **Guard 5 & Probe E (Ed25519-Signed Ledger, Multi-Parent Verification & Associative Merkle-DAG):** Every claim is signed with RFC 8032 Ed25519 (`test_28`), `verify_chain_list` rejects forged multi-parent entries not chained from `VRK_TRUST_ROOT_DIGEST` (`test_26`), and `merge_divergent_ledgers` satisfies $(A \sqcup B) \sqcup C = A \sqcup (B \sqcup C)$ and $A \sqcup A = A$ (Theorem `merkle_dag_semilattice_laws`, `test_19`, `test_20`, `test_25`).

References

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